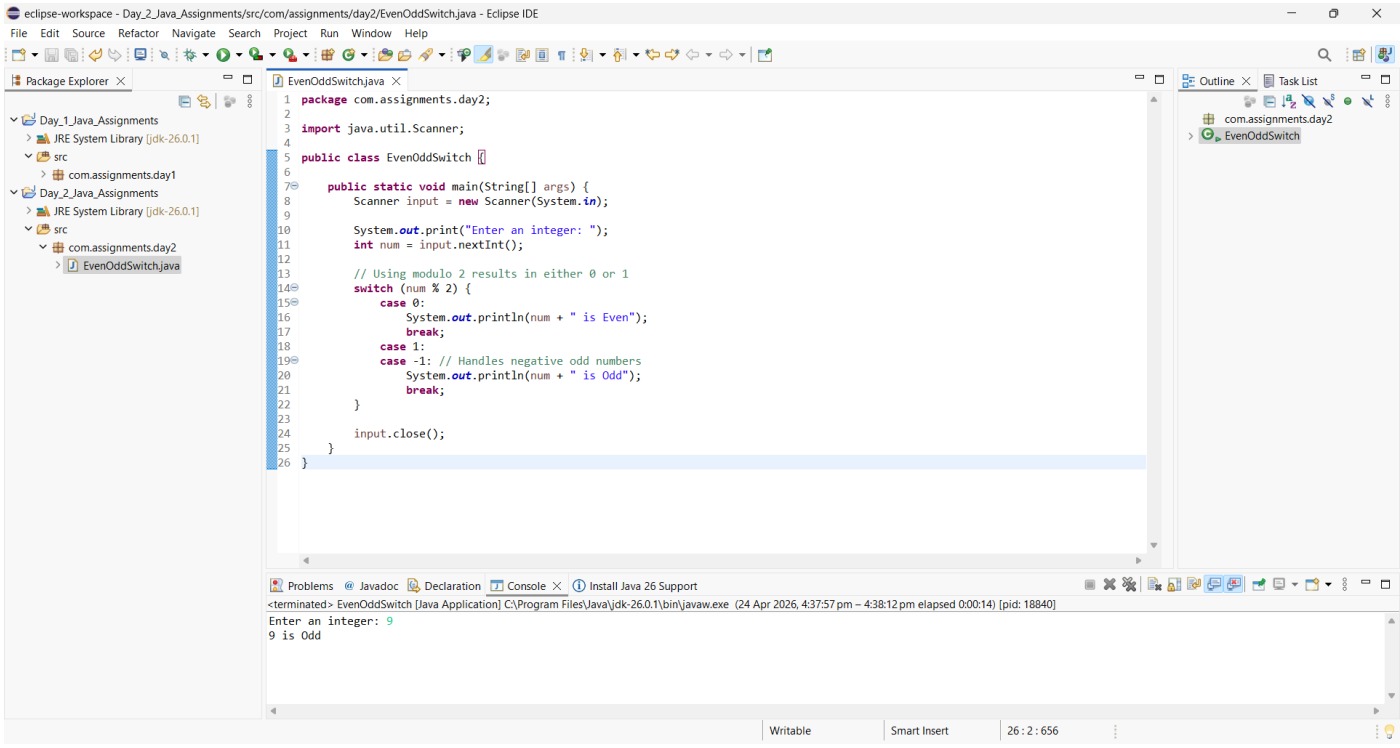
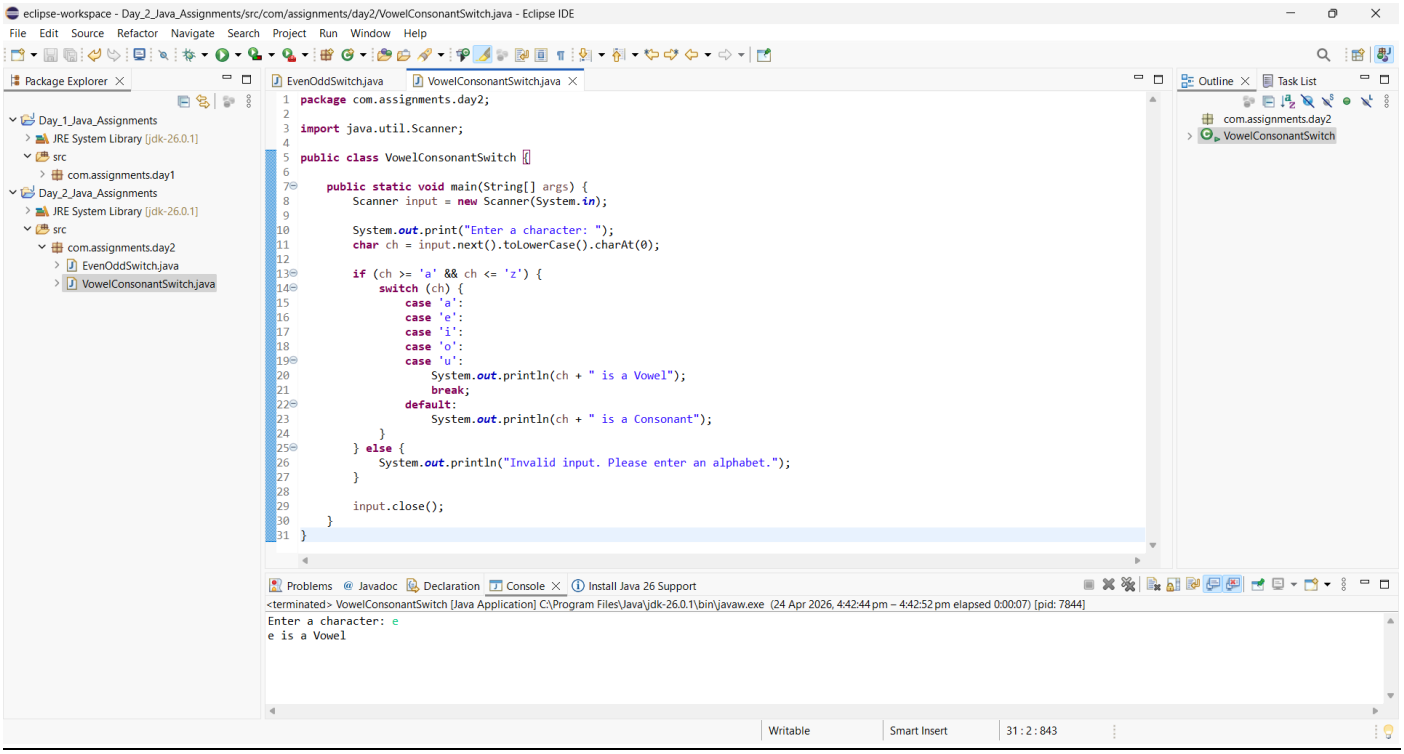
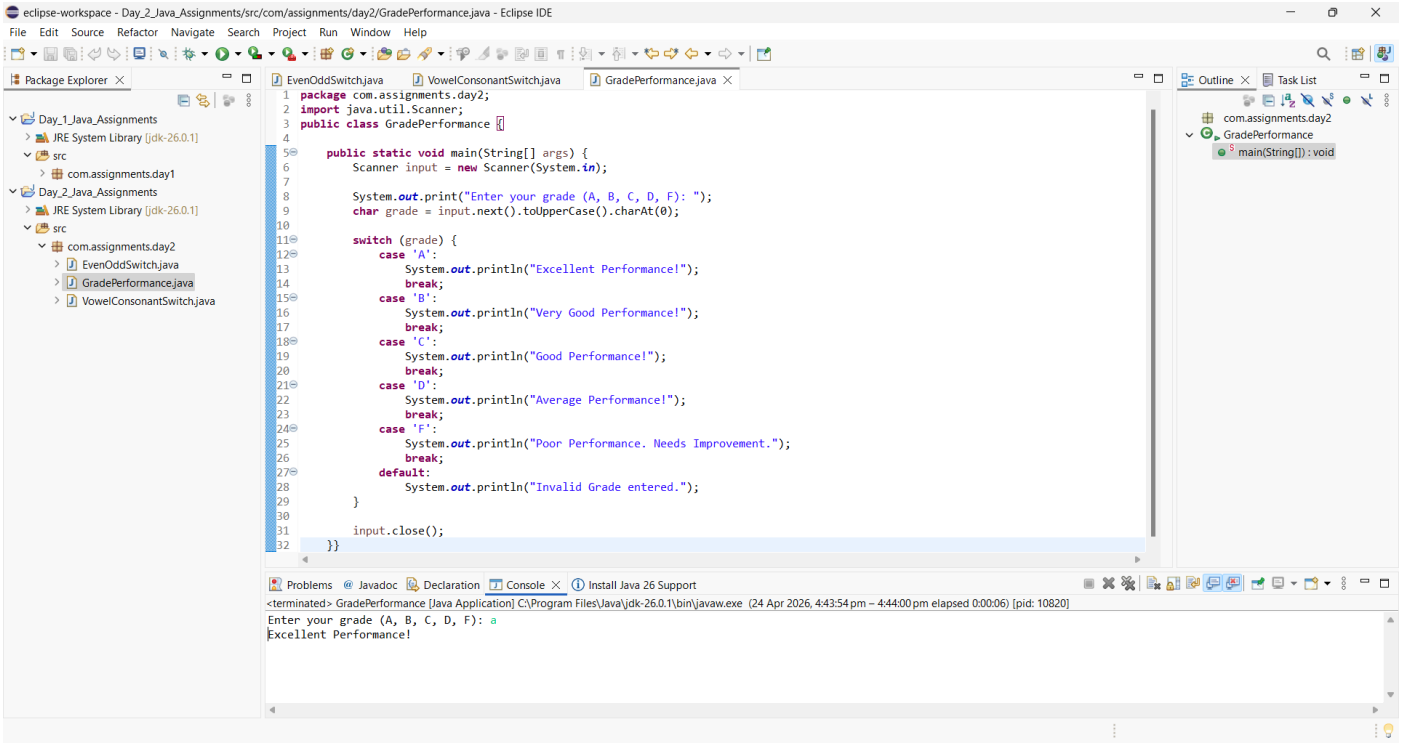
1. Write a program to check whether a number is even or odd using switch-case.




2. Write a program that takes a character as input and checks whether it is a vowel or consonant using switch-case.



3. <u>Write a program that takes a grade (A, B, C, D, F) and displays the performance message.</u>



4. <u>Create a menu-driven program for ATM operations: • 1 → Check Balance • 2 → Deposit • 3 → Withdraw • 4 → Exit Perform the operation based on user choice.</u>

eclipse-workspace - Day_2_Java_Assignments/src/com/assignments/day2/ATMOperations.java - Eclipse IDE

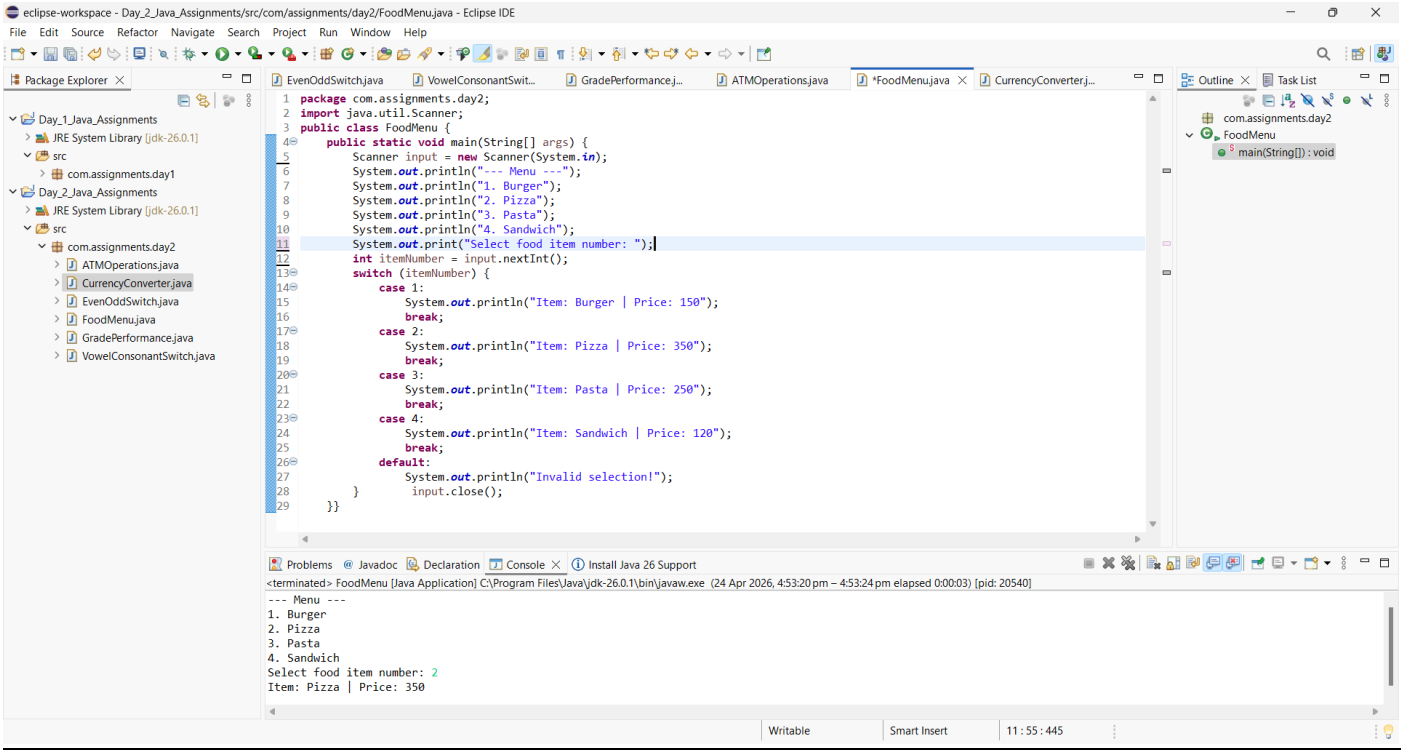
```java
package com.assignments.day2;

import java.util.Scanner;

public class ATMOperations {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        double balance = 5000.0;

        System.out.println("ATM Menu:");
        System.out.println("1 -> Check Balance");
        System.out.println("2 -> Deposit");
        System.out.println("3 -> Withdraw");
        System.out.println("4 -> Exit");
        System.out.print("Select an option: ");

        int choice = input.nextInt();

        switch (choice) {
            case 1:
                System.out.println("Current Balance: " + balance);
                break;
            case 2:
                System.out.print("Enter deposit amount: ");
                double deposit = input.nextDouble();
                balance += deposit;
                System.out.println("Successfully deposited. New Balance: " + balance);
                break;
            case 3:
                System.out.print("Enter withdrawal amount: ");
                double withdrawal = input.nextDouble();
```

eclipse-workspace - Day_2_Java_Assignments/src/com/assignments/day2/ATMOperations.java - Eclipse IDE

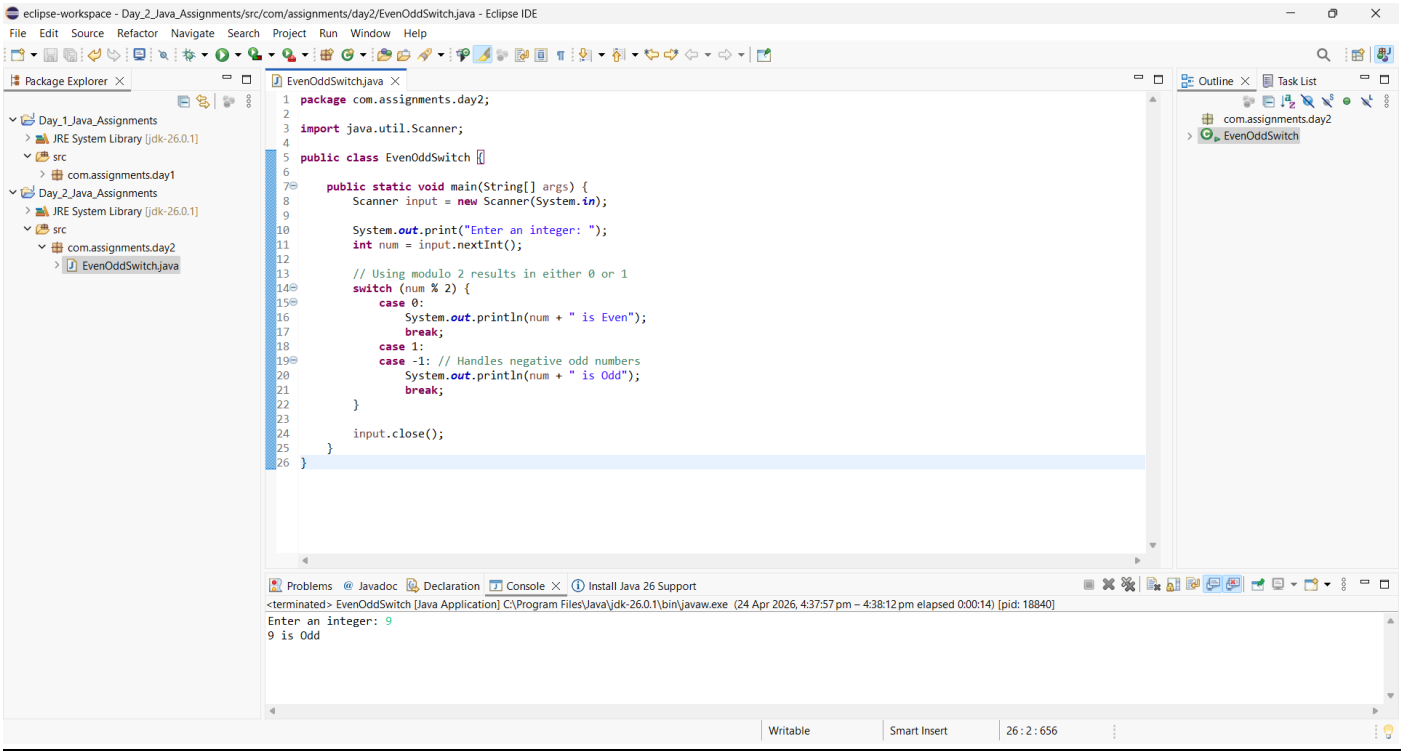
```java
        int choice = input.nextInt();

        switch (choice) {
            case 1:
                System.out.println("Current Balance: " + balance);
                break;
            case 2:
                System.out.print("Enter deposit amount: ");
                double deposit = input.nextDouble();
                balance += deposit;
                System.out.println("Successfully deposited. New Balance: " + balance);
                break;
            case 3:
                System.out.print("Enter withdrawal amount: ");
                double withdrawal = input.nextDouble();
                if (withdrawal <= balance) {
                    balance -= withdrawal;
                    System.out.println("Successfully withdrawn. Remaining Balance: " + balance);
                } else {
                    System.out.println("Insufficient balance!");
                }
                break;
            case 4:
                System.out.println("Exiting ATM. Thank you!");
                break;
            default:
                System.out.println("Invalid selection!");
        }

        input.close();
    }
}
```

```
<terminated> ATMOperations [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (24 Apr 2026, 4:45:11 pm – 4:45:25 pm elapsed 0:00:13) [pid: 17648]
3 -> Withdraw
4 -> Exit
Select an option: 2
Enter deposit amount: 1000
Successfully deposited. New Balance: 6000.0
```

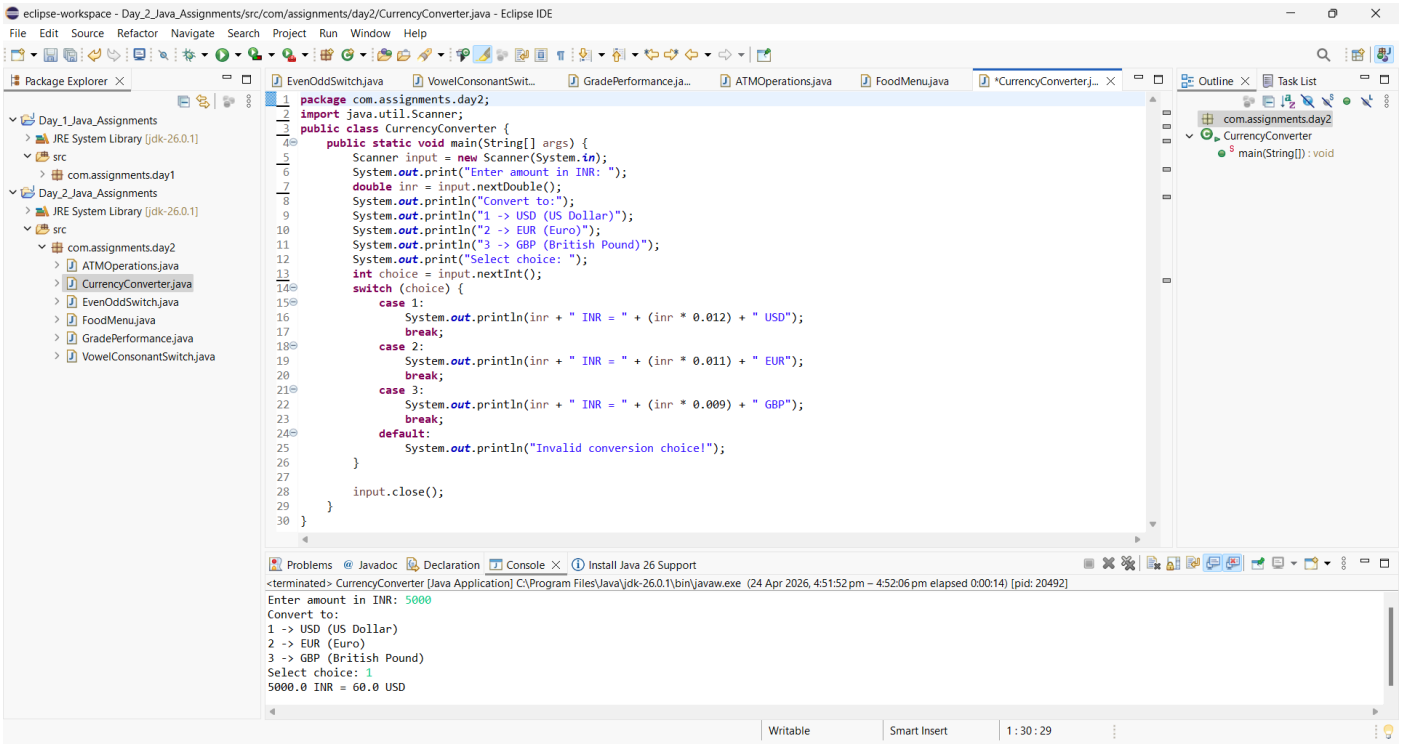
5. <u>Create a program where user selects food item number and the program displays item name and price.</u>



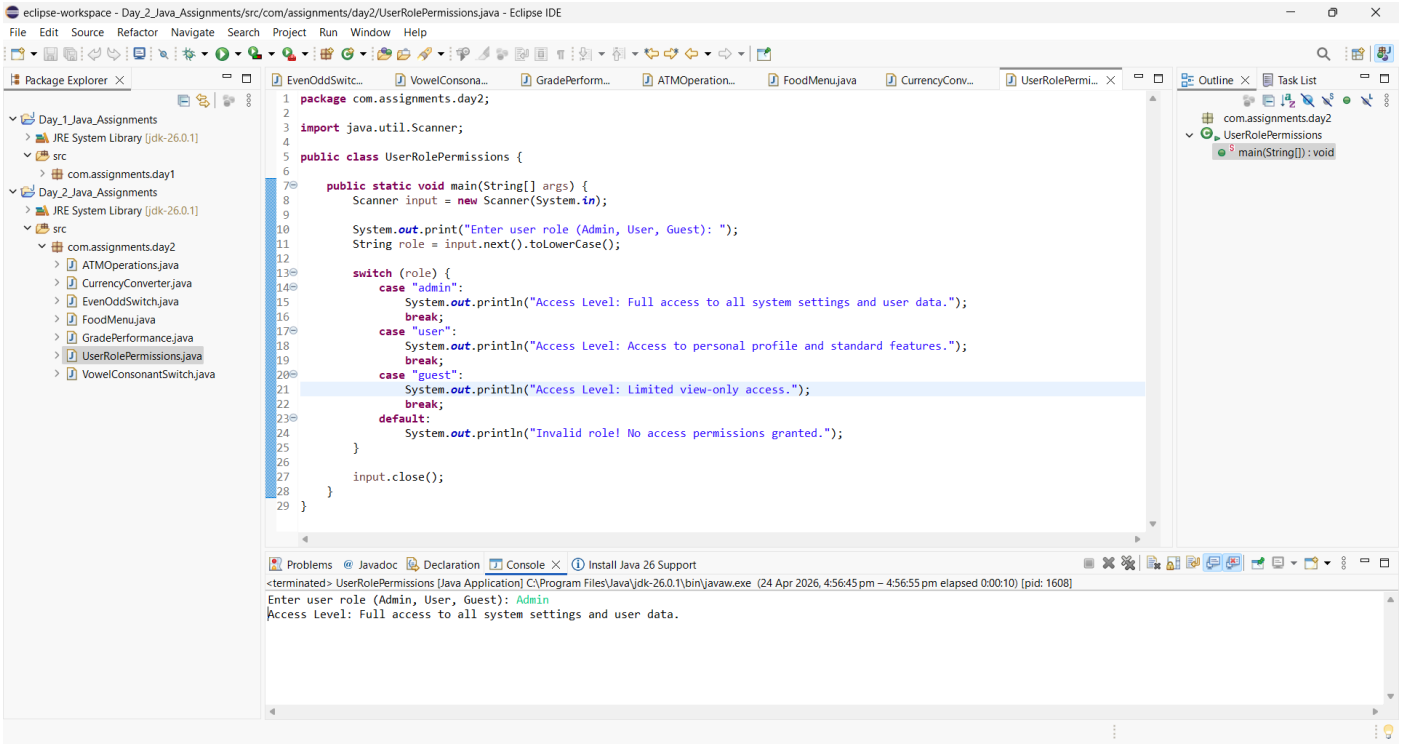

6. <u>Write a program to check whether a number is even or odd using switch-case.</u>



7. Write a program that converts currency based on user choice (e.g., INR to USD, INR to EUR, etc.).




8. Create a program that checks user role (Admin, User, Guest) and displays access permissions.



9. Write a program to calculate the factorial of a given number using a while loop.

eclipse-workspace - Day_2_Java_Assignments/src/com/assignments/day2/FactorialCalculator.java - Eclipse IDE

```java
package com.assignments.day2;

import java.util.Scanner;

public class FactorialCalculator {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = input.nextInt();

        int i = 1;
        long factorial = 1;

        if (num < 0) {
            System.out.println("Factorial is not defined for negative numbers.");
        } else {
            while (i <= num) {
                factorial *= i;
                i++;
            }
            System.out.println("Factorial of " + num + " is: " + factorial);
        }

        input.close();
    }
}
```

```
Enter a number: 8
Factorial of 8 is: 40320
```

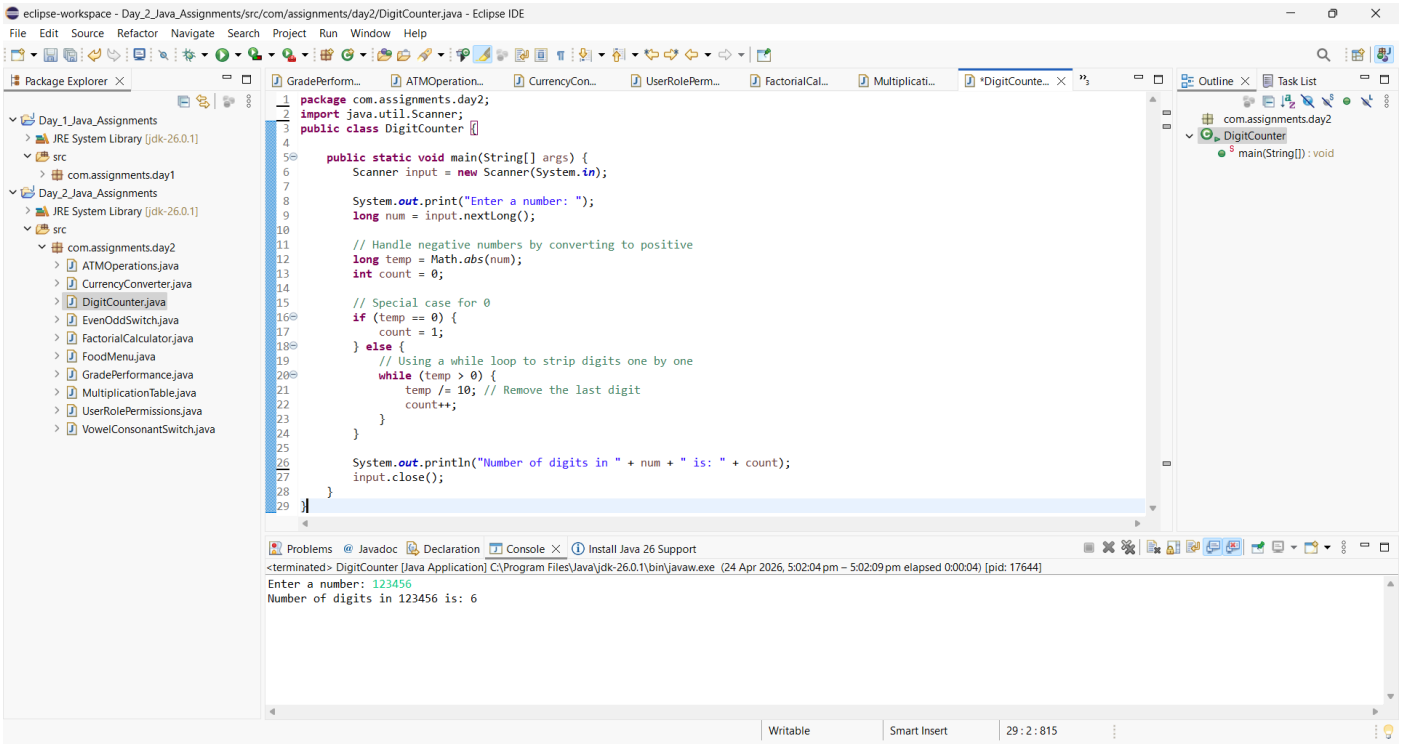
10. Write a program to print the multiplication table of a given number using a while loop.

eclipse-workspace - Day_2_Java_Assignments/src/com/assignments/day2/MultiplicationTable.java - Eclipse IDE

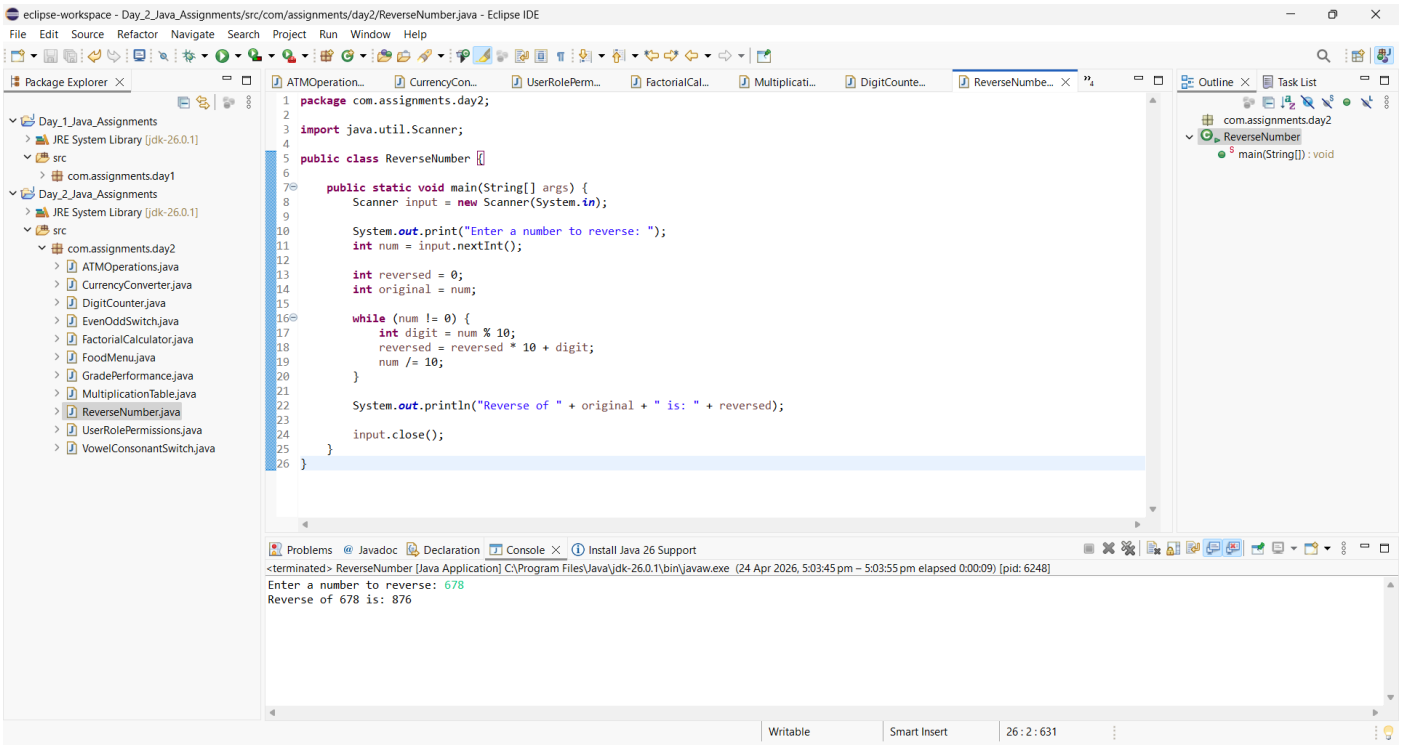
```java
package com.assignments.day2;

import java.util.Scanner;

public class MultiplicationTable {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = input.nextInt();

        int i = 1;

        System.out.println("Multiplication Table of " + num + ":");

        // Using a while loop to print the table up to 10
        while (i <= 10) {
            System.out.println(num + " x " + i + " = " + (num * i));
            i++;
        }

        input.close();
    }
}
```

```
Enter a number: 12
Multiplication Table of 12:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120
```

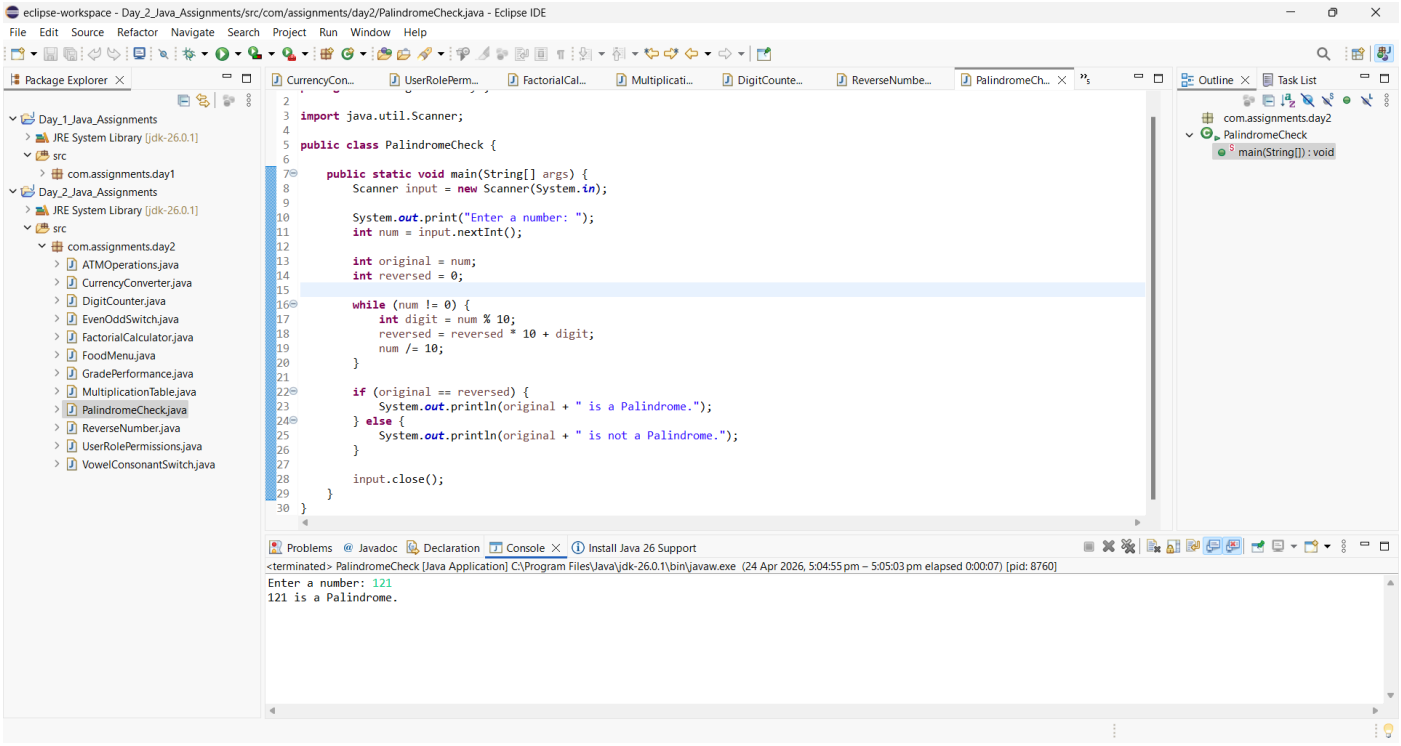
11. Write a program to count the number of digits in a given number using a while loop.



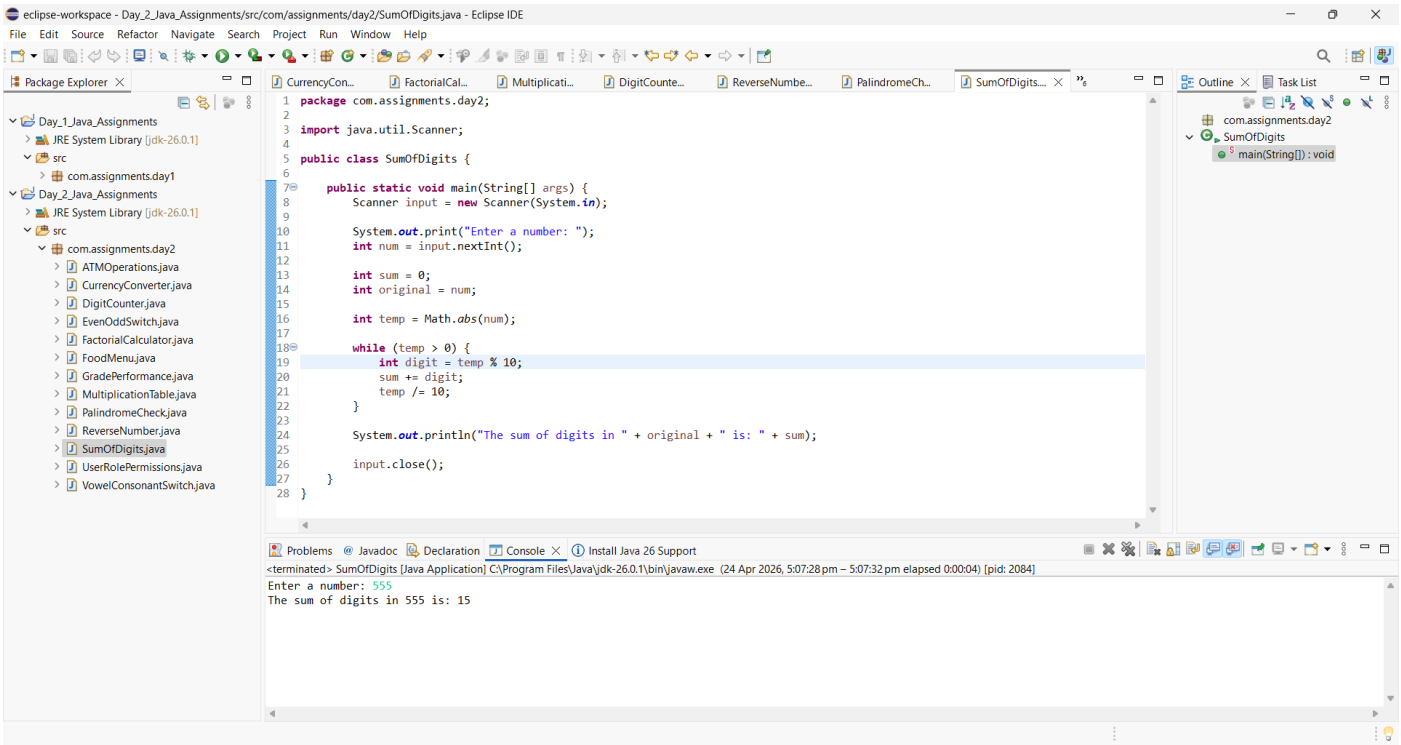

12. Write a program to reverse a given number using a while loop.



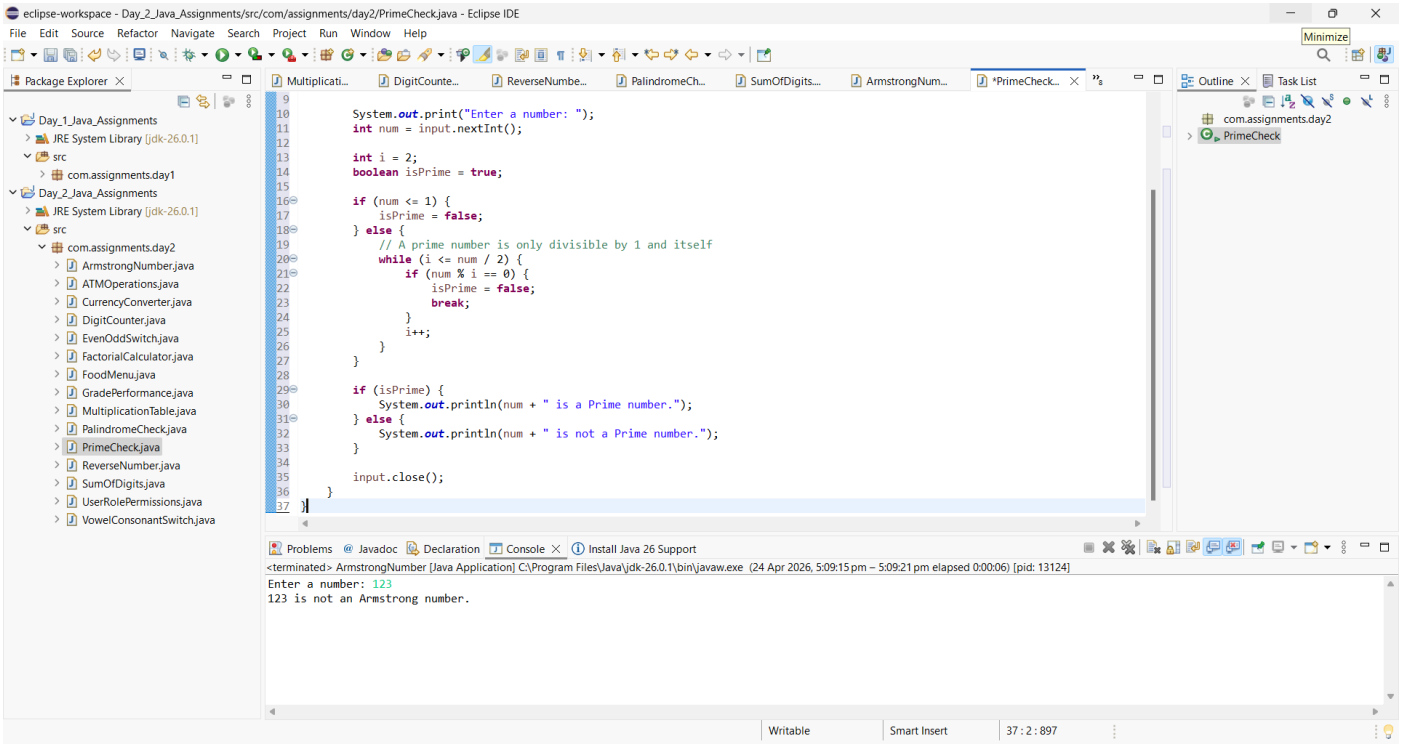
13. Write a program to check whether a number is a palindrome using a while loop.



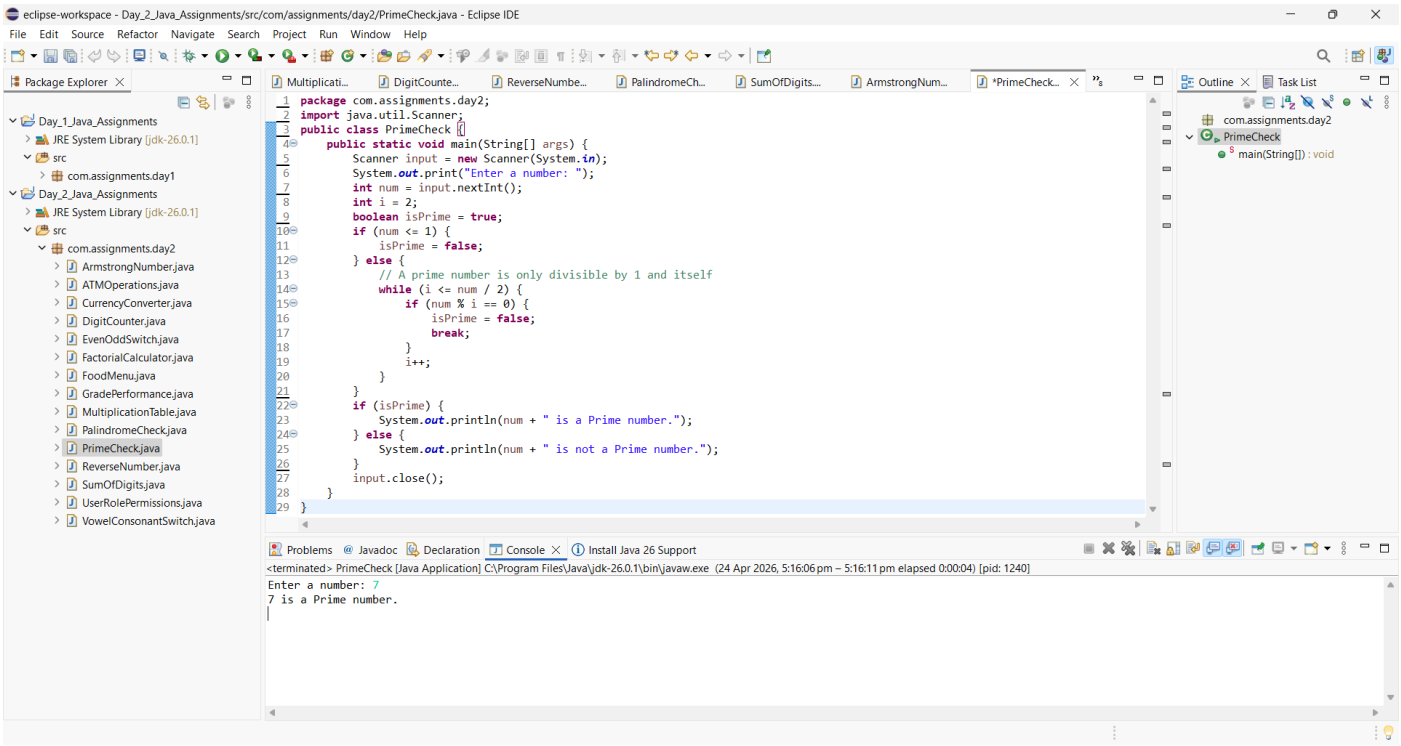
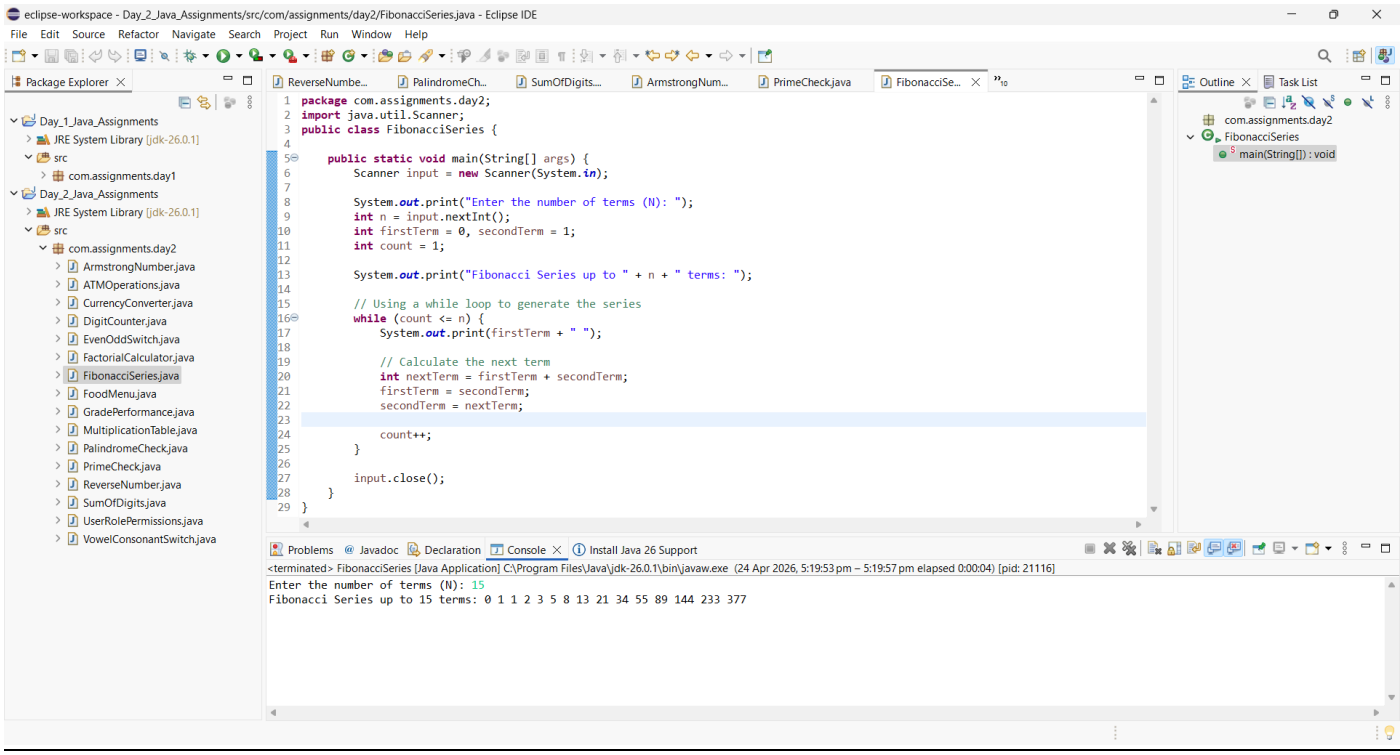

14. Write a program to find the sum of digits of a number using a while loop.



15. Write a program to check whether a number is an Armstrong number using a while loop.




16. Write a program to check whether a number is prime using a while loop.



17. <u>Write a program to print Fibonacci series up to N terms using a while loop.</u>